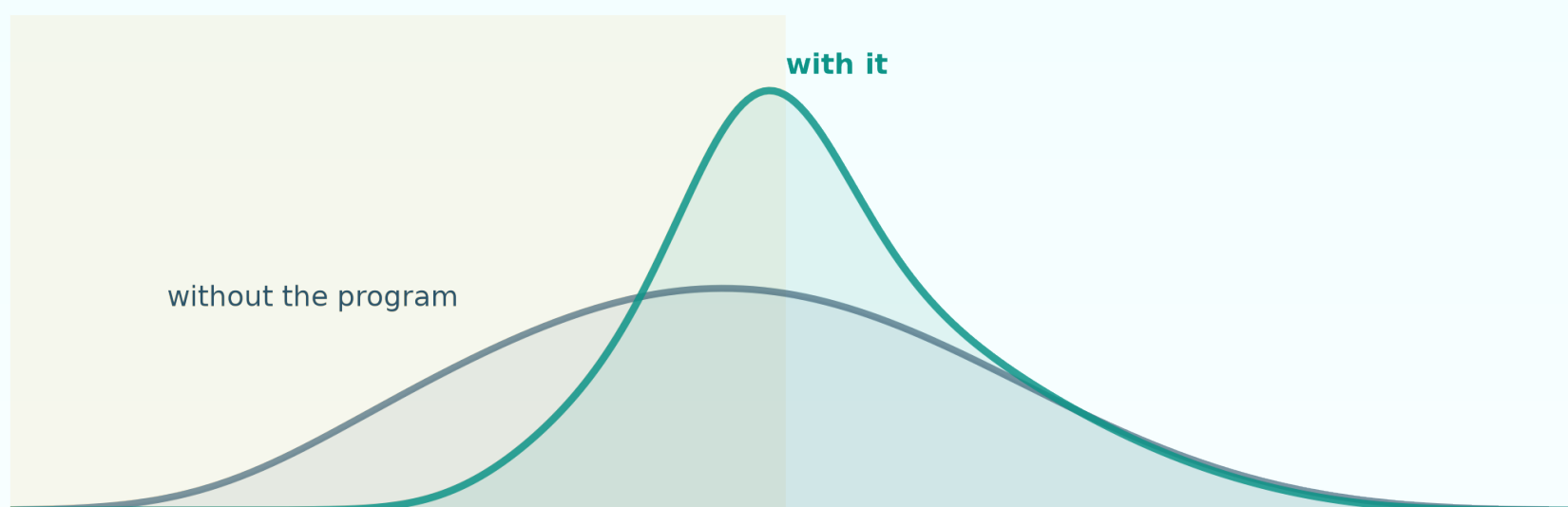


diff - diff

The effect your DiD can't see.



Real programs move parts of the distribution.

The mean averages them away.

ChangesInChanges: distributional DiD. Now in diff-diff.

Athey & Imbens (2006), Econometrica 74(2), 431-497.

THE PROBLEM

A real effect, averaged into nothing.

A loyalty program lifts low spenders by up to 65% and top spenders not at all.
(Simulated customer-spend 2x2, so the truth is known by construction.) Run the
standard playbook:

\$0.22

mean DiD, in dollars

p = 0.90 CI [-\$3.18, \$3.62]

verdict: kill the program

\$3.01

the TRUE mean effect

known by construction

The effect lives where
the mean isn't looking.

And it gets worse.

Same data. One change:

$\log(\text{spend})$.

+14.0%

$p = 0.001$

Your scale choice just became
your conclusion.

And neither verdict is right (truth: \$3.01). The simulated market trend is a power-law transform of spend - additive on neither scale across groups with different customer mixes.

Choosing a scale IS choosing the assumption.

Athey & Imbens (2006): compare ranks, not scales.

II

The assumptions of the proposed model are invariant to the scaling of the outcome.

- Athey & Imbens (2006), Econometrica 74(2), p. 431

Their Changes-in-Changes model swaps parallel trends in means for a rank story: a treated customer's counterfactual is the outcome move of a control customer at the same rank. No scale enters the assumptions - and none enters the answer.

The receipt (median counterfactual quantile):

\$20.3166

fit in dollars

\$20.3166

exp(fit in logs)

Identical across all 19 quantiles - max relative difference 0.00e+00 on this run (unconditional fits; means and ATTs stay scale-specific for every estimator).

One composition builds the whole counterfactual.

$$F_{11}^N(y) = F_{10}(F_{00}^{-1}(F_{01}(y)))$$

3. how many treated-pre customers sit below it

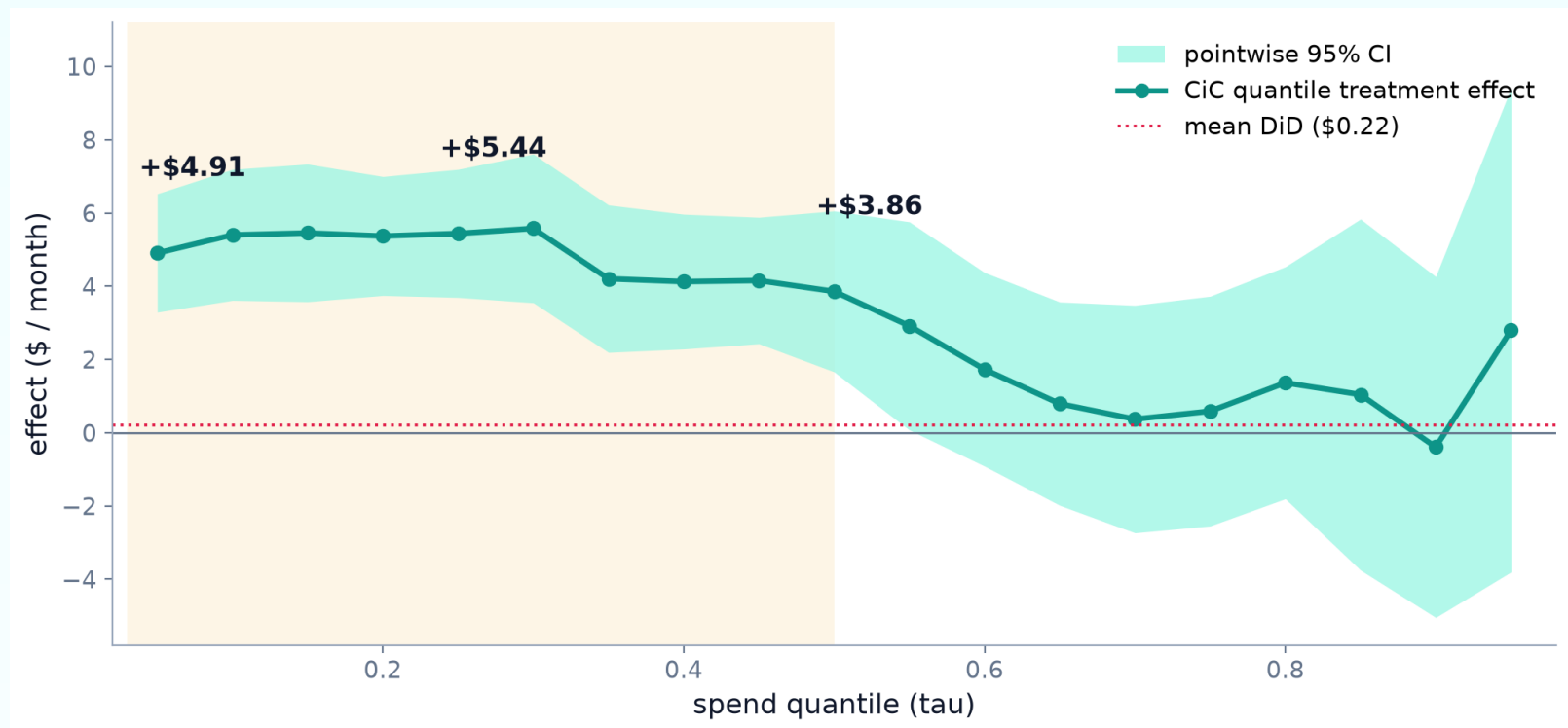
2. the same rank's outcome in control-pre

1. the rank of y among control-post outcomes

$$\text{QTE}(\tau) = F_{11}^{-1}(\tau) - F_{11}^{N, -1}(\tau)$$

Read inside-out: rank a treated outcome among control-post, walk that rank back to control-pre, and count the treated-pre customers below it. The result is the treated group's entire no-program outcome distribution (Theorem 3.1) - quantile treatment effects at every tau, and the ATT as its mean. Nonparametrically identified; panel or repeated cross-sections.

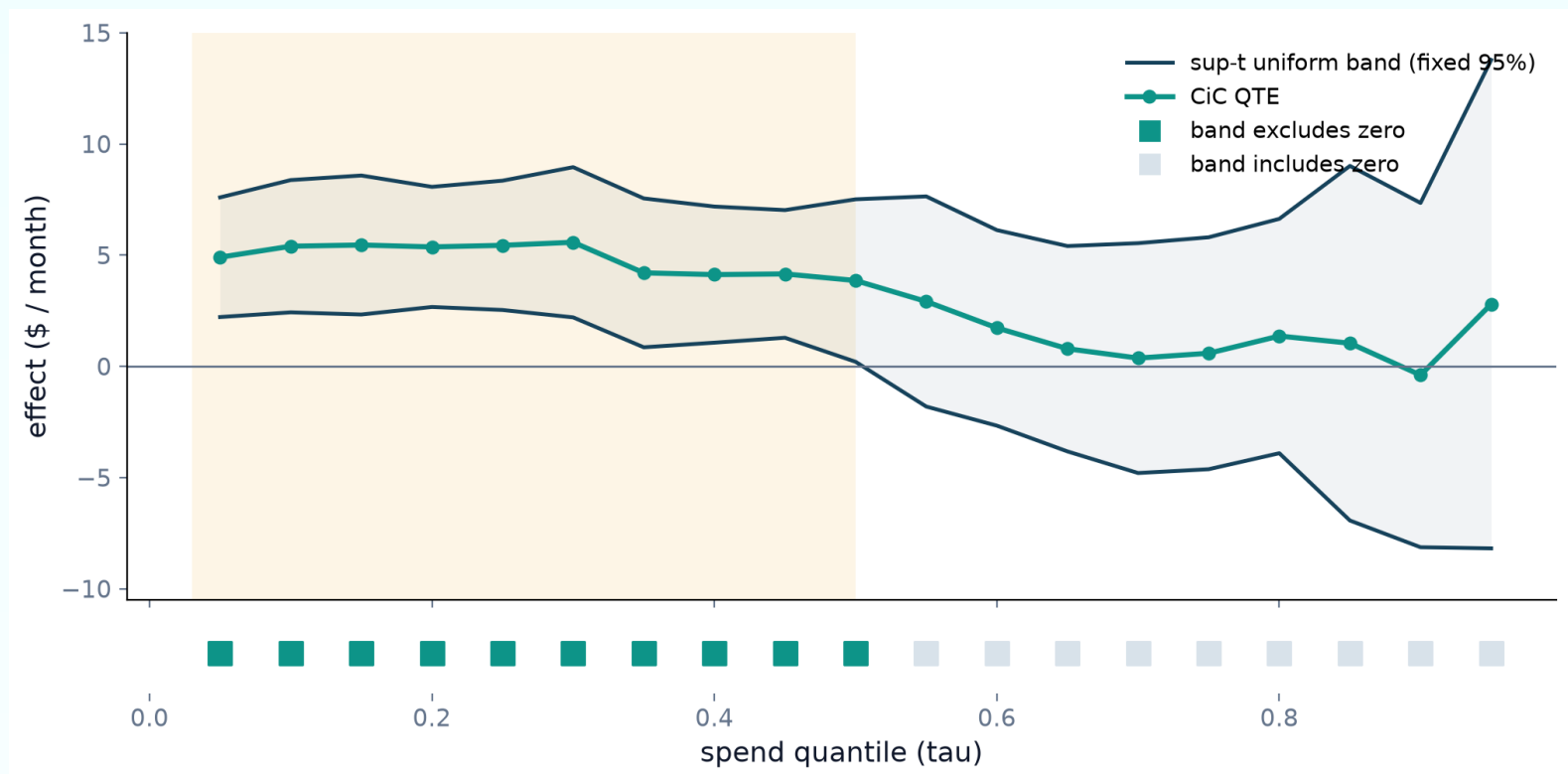
Same program. Now visible.



Gains of about +\$5/month across the bottom half of the spend distribution, fading to nothing distinguishable from zero above the 70th percentile. And the ATT - \$3.05 ($p = 0.004$) - lands on the \$3.01 truth the mean missed.

"The bottom half moved."

One test, nineteen quantiles.



Nineteen pointwise intervals over-reject when read together. Sup-t uniform bands (fixed 95%, a qte-parity convention) exclude zero for every quantile from 0.05 through 0.50 - and none above. The pointwise blip at $\tau = 0.55$ ($p = 0.044$) evaporates under the bands: exactly the trap they exist to prevent. Above the median the bands are silent, not exonerating.

Three lines to the whole distribution.

```
from diff_diff import ChangesInChanges

results = ChangesInChanges(n_bootstrap=999, seed=27).fit(
    df, outcome='spend', treatment='treated', time='post',
)

results.att          # $3.05 (p = 0.004)
results.quantile_effects  # QTE at 19 quantiles
results.uniform_bands()  # sup-t joint bands

# customer mix differs? fit(..., covariates=['tenure'])
# same units both periods? ChangesInChanges(panel=True)
```

Same sklearn-style API as every diff-diff estimator.

When the mix differs, condition.

If the treated group's mix differs on something that also drives the trend, the unconditional comparison is confounded. Here: treated customers skew longer-tenure, tenure drives engagement growth, and the true effect is 6.0 points - for everyone.

9.38

unconditional CiC

CI [8.32, 10.45]

excludes the truth:

confidently wrong

6.53

covariates=['tenure']

CI [5.35, 7.70]

covers the truth (6.0)

The conditional fit compares within covariate values: per-cell linear quantile regressions on a 99-tau grid, following R qte's xformula conventions verbatim - with a conditional support diagnostic standing guard.

Built for real pipelines.

Panel + repeated cross-sections

`panel=True` enforces balanced units +
unit-block bootstrap

Bootstrap + sup-t uniform bands

seeded, reproducible; fixed-95% bands (qte
parity)

Covariates

quantile-regression conditioning via
`covariates=[...]`

Loud guardrails

support, interior range + ties warnings; NaN,
never a guess

`practitioner_next_steps()`

distributional next-step guidance, built in

sklearn-style API

`fit()` / `summary()` / `to_dataframe()`, like every
estimator

Validated against R qte 1.3.1

Unconditional point estimates match the qte golden files to 1e-10; the type-1 quantile arithmetic is pinned bit-exactly (`atol=0`). Covariate branch: conventions exact given R inputs; the LP solver is proven optimal at every τ .

diff - diff

Ask your 2x2 who moved -
not just how much on average.

```
pip install diff-diff
```

Want the full walkthrough? Tutorial 27 in the docs reproduces this example end to end -
and every number in this deck is locked by drift tests in CI.

github.com/igerber/diff-diff

diff-diff.readthedocs.io

ChangesInChanges - Athey & Imbens (2006). Now in diff-diff.